DM42 PROBABILITY, STATISTICS & REGRESSION

From Counting and Chance to Measurement, Data Analysis and Prediction

A patient, practical keystroke handbook

Second edition • optional refreshers • complete worked examples

Welcome

Probability and statistics have a reputation for speaking a private language. A formula may contain an unfamiliar Greek letter, a vertical bar, a symbol that looks like an upside-down U, and several subscripts—all before anyone has explained what question is being answered. That is not a failure on the reader's part. It is a teaching problem.

This handbook starts with the questions and translates the language as it goes. Every important expression is presented in three layers:

1. the mathematical notation;
2. how to read it aloud;
3. what it means in the situation at hand.

Only then do we work through the calculator procedure. The purpose of the DM42 is not to hide the reasoning. It is to handle the repetitive arithmetic after we have decided what the arithmetic means.

Two routes through the book

Guided route. Read the optional refresher boxes. They review RPN, powers, percentages, logarithms, summation signs, graphs and other ideas that probability and statistics use. No previous course in statistics is assumed.

Direct route. Skip the refresher boxes when the material is familiar. The main worked examples remain complete, so the argument will still make sense.

Both routes lead through practical problems: inspection, reliability, communications, measurement, calibration, delivery planning, inventory and physical experiments. Each main example ends with an interpretation and a check, because a number without a meaning is not yet an answer.

Keystroke language

Boxed labels represent keys or on-screen softkeys. Read the sequence from left to right:

8 ENTER 3 f × COMB

The sequence enters 8 and 3, opens the probability menu with the shifted multiply key, and selects COMB. It produces 56.

Notation	Meaning
f	yellow shift key
STAT	top-row key carrying the yellow STAT legend
×	physical multiply key
COMB	menu label shown above a softkey
↓	move to the next menu row
→ 56	the display should show 56

The calculator uses Reverse Polish Notation (RPN). When a two-number operation is performed, the lower stack value is normally called y and the displayed value is x. For combinations and permutations, enter the total n, press ENTER, enter the selected number r, then choose the function.

Optional refresher: the four-level stack

Think of the stack as a short pile of values. Typing a number puts it in X. ENTER copies it upward and prepares X for the next number. A two-number operation uses Y and X, places its result in X, and lets the remaining values drop. When a sequence feels uncertain, write the intended Y and X values beside the keys.

What is assumed

The reader is expected to be comfortable entering ordinary numbers and the four arithmetic operations. Everything else needed for the worked examples is introduced or refreshed. The book distinguishes a mathematical assumption from a calculator fact: independence, for example, is a claim about the real system, not a setting on the calculator.

Rounding

The examples retain the calculator's internal precision and round only the reported answer. Four to six displayed decimals are usually enough for inspection. If a value will be used in another calculation, keep it on the stack or in a register instead of retyping the rounded display.

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Part I

Learning the Language

CHAPTER 1

Outcomes, events and the questions we ask

Probability begins with uncertainty, but it does not begin with formulas. It begins by naming what can happen.

Suppose a technician selects one device from a shipment and tests it. The individual possibilities—pass, fail, or perhaps "test inconclusive"—are *outcomes*. An *event* is one outcome or a collection of outcomes that answers a question. "The device fails" is an event. So is "the device does not pass on its first test."

We usually name events with capital letters. If F means "the device fails," then

$$P(F)$$

means the probability that the event F occurs.

Read the mathematics

Read $P(F)$ as "the probability of F." If the context says that F means failure, it is equally natural to read it as "the probability of failure."

1.1 A probability is a proportion on a 0-to-1 scale

A probability of 0 means impossible; 1 means certain. A probability of 0.25 means 25%, or about one occurrence in four comparable opportunities over the long run. The three forms below describe the same value:

$$0.25 = 25\% = \frac{1}{4}.$$

Optional refresher: percentages

To change a probability into a percentage, multiply by 100. To change a percentage into a probability, divide by 100. Thus 2.5% becomes $2.5/100 = 0.025$.

DM42 keystrokes

2.5 ENTER 100 ÷ → 0.025

1.2 Sets are simply collections

Probability borrows symbols from set theory because an event is a collection of outcomes. Let A mean "the device triggers an alarm" and D mean "the device is defective."

1.2.1 Intersection: both conditions

$$A \cap D$$

Read the mathematics

Read this as "A intersection D," or more plainly, "A and D." The upside-down U symbol $\cap$ means both conditions are true.

Here it means that the device is defective *and* the test triggers an alarm.

1.2.2 Union: either condition or both

$$A \cup D$$

Read the mathematics

Read this as "A union D," or "A or D." In probability, *or* normally includes the possibility that both occur.

1.2.3 Complement: the event does not occur

$$D^c$$

Read this as "D complement" or "not D." It contains every outcome outside D. If D means defective, D^c means not defective.

1.3 The vertical bar: restrict the conversation

The expression

$$P(D \mid A)$$

is read "the probability of D given A." The vertical bar means that we restrict attention to cases in which A is already known to have occurred. Here the question is not "How many devices are defective?" It is "Among the devices that triggered an alarm, how many are actually defective?"

Interpretation

The phrase after the bar is the information we have. The event before the bar is what we want to know. Read the right-hand side first in ordinary language: "Given an alarm, what is the chance of a defect?"

1.4 Random variables turn outcomes into numbers

Let X be the number of failed devices in a sample of 20. X is a *random variable*: a rule that assigns a number to the outcome of an uncertain process. Statements about X describe ranges of possible counts.

Expression	Read and interpret
$P(X = 2)$	probability of exactly two failures
$P(X \leq 2)$	probability of at most two: 0, 1 or 2
$P(X < 2)$	probability of fewer than two: 0 or 1
$P(X \geq 2)$	probability of at least two: 2, 3, 4, . . .
$P(1 < X \leq 4)$	probability of more than one but no more than four: 2, 3 or 4

Words such as *at least* and *at most* include the boundary. This small language point prevents many probability errors.

What to carry forward

Before calculating, translate the question into a sentence about outcomes. Then translate that sentence into symbols. If the sentence and symbols do not say the same thing, stop before touching the calculator.

1.5 Reading formulas without being intimidated

A formula is a compressed instruction. It names quantities and specifies how they are related. You are allowed to unpack it slowly.

1.5.1 Letters, subscripts and hats

The same few conventions appear throughout the book.

Notation	Meaning
x_i	the ith observed x value; x_3 is the third value
n	the number of observations or trials
p	a probability of success on one trial
k	a particular count being considered
$\bar{x}$	“x-bar,” the mean of a sample
μ	“mu,” the mean of a population or model
s	sample standard deviation
σ	“sigma,” population or model standard deviation
$\hat{y}$	“y-hat,” a fitted or predicted value of y

A hat means estimated or predicted. A bar means averaged. A subscript identifies one member of a sequence; it is not multiplication.

1.5.2 The summation sign

$$\sum_{i=1}^{n} x_i$$

Read the mathematics

Read this as “the sum of x sub i, from i equals one to n.” In plain language: add every observed x value from the first through the nth.

For $x_1 = 4$, $x_2 = 7$ and $x_3 = 9$,

$$\sum_{i=1}^{3} x_i = 4 + 7 + 9 = 20.$$

The DM42 statistics system performs this repeated addition when you press [Σ+]. The Greek capital sigma Σ is the standard symbol for a sum.

1.5.3 Products, powers and factorials

$$\prod_{i=1}^{n} R_i$$

uses capital pi $\prod$ to mean a product: multiply all the indicated values. It will appear in series reliability.

The expression p^k means multiply p by itself k times. On the calculator, enter the base, then the exponent, then [y^x]:

DM42 keystrokes

For 0.98^9: [.98] [ENTER] [9] [y^x] → 0.83374776

A factorial is

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1,$$

with $0! = 1$. It counts arrangements and appears inside several probability formulas.

DM42 keystrokes

For 6!: [6] [f] [×] [N!] → 720

1.5.4 The exponential and the natural logarithm

The number $e \approx 2.71828$ is the natural base for continuous growth and decay. The expression $e^{-2.4}$ is read "e to the power negative 2.4."

DM42 keystrokes

[2.4] [+/−] [e^x] → 0.09071795

The natural logarithm $\ln x$ reverses the exponential: if $e^a = x$, then $\ln x = a$. It appears in exponential, logarithmic and power regression.

Optional refresher: why logarithms help

A curved exponential relationship becomes a straight line after taking a logarithm. For $y = be^{mx}$,

$$\ln y = mx + \ln b.$$

The calculator uses this transformed straight line to fit the exponential model, then converts the intercept back to b.

1.5.5 An equality is a claim; an approximation is a report

The equals sign $=$ states exact equality within the mathematical model. The approximation sign $\approx$ announces rounding or an approximate model. For example,

$$\frac{1}{3} = 0.3333\ldots \approx 0.3333.$$

Measured-data equations are fitted descriptions, so it is often more honest to write $\hat{y} = mx + b$ than $y = mx + b$.

1.6 From a question to a calculator plan

The calculator cannot decide which probability model describes a situation. Use this five-step conversation before calculation.

1.6.1 Step 1: identify the quantity

Is the answer a count, a probability, an average, a spread, a predicted measurement or an expected cost? Write the answer's units. A probability has no physical unit; an expected cost has a currency unit; a fitted slope has response units per predictor unit.

1.6.2 Step 2: define the event or variable

Write one sentence such as

$$X = \text{number of failed components in the sample of 20.}$$

This prevents X from quietly changing meaning halfway through a solution.

1.6.3 Step 3: name the assumptions

Typical assumptions include:

- trials are independent;
- the success probability is constant;
- sampling is without replacement;
- the event rate is stable during the interval;
- a straight line is a sensible description over the measured range;
- the sample represents the population of interest.

An assumption is not necessarily true. It is a condition under which the calculation answers the intended question.

1.6.4 Step 4: choose the shortest correct expression

"At least one" is usually shorter through the complement. A long binomial upper tail may be shorter as one minus a lower tail. A regression forecast inside the observed range is more defensible than one far outside it.

1.6.5 Step 5: predict the rough scale

Before calculating, ask what would be surprising. If a defect probability is 2.5% in a sample of 20, the expected count is $20(0.025) = 0.5$. A result claiming that ten defects are typical would fail immediately.

Optional refresher: calculator display versus stored precision

If the display is set to four decimal places, a value shown as 0.2213 may still be stored internally as 0.2212770889. Continue with the stored value. Do not type 0.2213 back into the calculator unless the problem explicitly requires that rounding.

What to carry forward

The reliable order is: question $\rightarrow$ meaning $\rightarrow$ assumptions $\rightarrow$ expression $\rightarrow$ keystrokes $\rightarrow$ interpretation. Starting with a menu usually makes the method harder to understand.

Part II

Counting and Foundational Probability

CHAPTER 2

Counting: does order matter?

Counting problems often sound simple until the list becomes too long to write. The central question is whether two selections containing the same objects in a different order count as different results.

2.1 Anchor example: assigning inspection positions

Eight technicians are available for three distinct positions: lead inspector, recorder and verifier. How many assignments are possible if no person holds more than one position?

Why this method?

The positions are different. Alice as lead and Ben as recorder is not the same assignment as Ben as lead and Alice as recorder. Order matters, so this is a permutation.

The permutation formula is

$$ {}_nP_r = \frac{n!}{(n-r)!}. $$

Read the mathematics

Read ${}_nP_r$ as "the number of permutations of n objects taken r at a time." Here $n = 8$ available technicians and $r = 3$ positions.

DM42 keystrokes

8 ENTER 3 f × PERM

Display: **336**

Interpretation

There are 336 distinct ways to fill the three named positions. The result counts assignments, not groups of people.

Sanity check

There are 8 choices for lead, 7 remaining choices for recorder and 6 for verifier. $8 \times 7 \times 6 = 336$, which independently confirms the result.

2.2 Variation: selecting the inspection team

Suppose the same manager only wants to choose three technicians; their roles will be assigned later. Now Alice, Ben and Cara form the same team regardless of the order in which their names are written. Order does not matter.

The combination formula is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

DM42 keystrokes

8 ENTER 3 f × COMB
Display: **56**

Each team can be assigned to the three positions in $3! = 6$ ways, so $56 \times 6 = 336$. This relation connects the two answers.

2.3 Variation: arranging all six instruments

Six distinct instruments can be inspected in

$$6! = 720$$

orders.

DM42 keystrokes

6 f × N! $\rightarrow 720$

Take care

Words such as *choose*, *committee* and *sample* often suggest combinations. Words such as *arrange*, *rank*, *route* and named positions often suggest permutations. The real test is not the word; it is whether changing the order creates a different outcome.

CHAPTER 3

Adding, multiplying and taking complements

The basic rules of probability are short, but choosing the correct rule depends on the story.

3.1 Mutually exclusive alternatives: add

If two events cannot occur together, their probabilities add. A selected component cannot simultaneously be graded A and rejected. If

$$P(A) = 0.72, \qquad P(R) = 0.03,$$

then

$$P(A \cup R) = 0.72 + 0.03 = 0.75.$$

Read the mathematics

Read $A \cup R$ as "A or R." Because the categories cannot overlap, no outcome is counted twice.

DM42 keystrokes

.72 ENTER .03 + $\rightarrow 0.75$

3.2 Overlapping events: subtract the overlap

When events can occur together,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Suppose 40% of service calls involve power, 30% involve communications and 12% involve both. Then

$$P(\text{power or communications}) = 0.40 + 0.30 - 0.12 = 0.58.$$

DM42 keystrokes

.40 ENTER .30 + .12 - $\rightarrow 0.58$

The overlap is subtracted because it was included once in each of the first two percentages.

3.3 Independent stages: multiply

Three independent series components have reliabilities 0.98, 0.97 and 0.995. The system works only if every component works:

$$P(\text{system works}) = 0.98(0.97)(0.995) = 0.945847.$$

DM42 keystrokes

[.98] [ENTER] [.97] [×] [.995] [×]
Display: **0.945847**

Interpretation

About 94.585% of comparable operating missions should succeed under the model. The arithmetic assumes that one component's failure does not change the others' failure probabilities.

3.4 The complement: everything else

The complement rule is

$$P(A^c) = 1 - P(A).$$

Two independent parallel units each have reliability 0.92. The system fails only if both units fail. One unit fails with probability $1 - 0.92 = 0.08$, so

$$P(\text{both fail}) = 0.08^2 = 0.0064,$$

and

$$P(\text{system works}) = 1 - 0.0064 = 0.9936.$$

DM42 keystrokes

[1] [ENTER] [.92] [-] [x^2] [1] [$x \leftrightarrow y$] [-]
Display: **0.9936**

Variation– at least one failure

Five independent components each succeed with probability 0.99. The chance of at least one failure is

$$1 - (0.99)^5 = 0.04900995.$$

DM42 keystrokes

[.99] [ENTER] [5] [y^x] [1] [$x \leftrightarrow y$] [-] $\rightarrow 0.04900995$

Although each component is 99% reliable, the five-component mission has about a 4.90% chance of at least one failure.

CHAPTER 4

Conditional probability and diagnostic evidence

Conditional probability changes the reference group. The formula is

$$P(D \mid A) = \frac{P(D \cap A)}{P(A)}.$$

Read the mathematics

Read this as "the probability of D given A equals the probability of D and A divided by the probability of A." The denominator says which cases remain in the conversation.

4.1 Anchor example: an inspection alarm

Only 1% of incoming units are actually defective. An automated screen alarms on 95% of defective units, but it also alarms on 5% of good units. A unit has just triggered an alarm. What is the probability that it is defective?

Let D mean defective and A mean alarm. The given information is

$$P(D) = 0.01, \quad P(A \mid D) = 0.95, \quad P(A \mid D^c) = 0.05.$$

Imagine 10,000 units. About 100 are defective and 9,900 are good.

- Defective units producing alarms: $100(0.95) = 95$.
- Good units producing false alarms: $9,900(0.05) = 495$.
- Total alarms: $95 + 495 = 590$.

Among those 590 alarms, only 95 indicate a real defect:

$$P(D \mid A) = \frac{95}{590} = 0.16101695.$$

DM42 keystrokes

Numerator $P(D \cap A)$: [.01] [ENTER] [.95] [×] [STO] [0] [0]
False-alarm contribution: [.99] [ENTER] [.05] [×]
Add the true-alarm contribution and divide: [RCL] [0] [0] [+] [RCL] [0] [0] [$x \leftrightarrow y$] [÷]
Display: **0.16101695**

Interpretation

An alarm raises the defect probability from 1% to about 16.1%, but it does not make a defect certain. False alarms dominate because good units are far more common to begin with.

Sanity check

The answer must be greater than the original 1% because an alarm is evidence of a defect. It must also be well below 95%, because 95% is the screen's sensitivity, not the probability of a defect after an alarm.

4.2 Independence is a statement about information

Events A and B are independent when

$$P(A \mid B) = P(A).$$

Learning that B occurred does not change the probability of A. Independence is different from mutual exclusivity. Two mutually exclusive events cannot occur together and are usually dependent: learning that one occurred tells us the other did not.

CHAPTER 5

Expected value: the long-run centre of a decision

Expected value combines possible outcomes with their probabilities:

$$E(X) = \sum_i p_i x_i.$$

Read the mathematics

Read this as "the expected value of X is the sum of each probability times its corresponding value." It is a probability-weighted average, not necessarily an outcome that can occur on one trial.

5.1 Anchor example: warranty provision

A product has the following one-year warranty outcomes:

Outcome	Probability	Cost
No claim	0.89	R0
Minor repair	0.07	R120
Major repair	0.03	R450
Replacement	0.01	R1,200

The expected direct cost is

$$E(C) = 0.89(0) + 0.07(120) + 0.03(450) + 0.01(1200).$$

DM42 keystrokes

.07 ENTER 120 × .03 ENTER 450 × + .01 ENTER 1200 × +

Display: **33.9**

Interpretation

The expected direct warranty cost is R33.90 per unit across a large portfolio. No individual claim costs R33.90. Most units cost nothing, while a few cost much more.

Variation– a fair die

For a fair die, each face has probability $1/6$:

$$E(X) = \frac{1+2+3+4+5+6}{6} = 3.5.$$

DM42 keystrokes

1 ENTER 2 + 3 + 4 + 5 + 6 + 6 ÷ $\rightarrow 3.5$

The expected value 3.5 is not a possible die result. It is the centre of many rolls.

Take care

Expected value is not automatically the correct decision criterion. Cash limits, risk, rare catastrophic losses and uncertainty in the probabilities may matter. It is the starting point for a transparent decision, not a substitute for judgement.

Part III

Probability Distributions and Simulation

CHAPTER 6

Distributions and the binomial model

A probability distribution lists the possible values of a random variable and the probability attached to each one. It is a complete map of the uncertainty.

6.1 A small example: two component tests

Suppose each tested component independently passes with probability 0.8. Let X be the number of passes in two tests. Then X can be 0, 1 or 2.

x	$P(X = x)$	Meaning
0	$(0.2)^2 = 0.04$	both fail
1	$2(0.8)(0.2) = 0.32$	one passes, one fails
2	$(0.8)^2 = 0.64$	both pass

The probabilities add to 1 because the table covers every possible value of X.

6.2 The expected value and spread of a distribution

The expected value is

$$E(X) = \sum_x xP(X = x).$$

For the two-test example,

$$E(X) = 0(0.04) + 1(0.32) + 2(0.64) = 1.6.$$

Across many pairs of tests, the average number of passes approaches 1.6.

The variance measures squared distance from the mean:

$$\mathrm{Var}(X) = \sum_x (x - \mu)^2 P(X = x),$$

and the standard deviation is $\sigma = \sqrt{\mathrm{Var}(X)}$. These model quantities use μ and σ; sample statistics later use $\bar{x}$ and s.

6.3 Choose a distribution by the experiment

Model	Recognising features	Typical question
Binomial	fixed number of independent yes/no trials; constant p	failures among 20 units
Hypergeometric	sample without replacement from a finite lot	defects found in five draws
Poisson	event count in a fixed interval at a stable average rate	alarms during one hour
Normal	continuous measurement clustered symmetrically around a centre	fraction inside tolerance

The distribution is chosen from how the data arise, not from the shape of a desired answer.

6.4 Binomial probability: repeated yes/no trials

A binomial experiment has four ingredients:

1. a fixed number n of trials;
2. two outcomes on each trial;
3. the same success probability p on every trial;
4. independence between trials.

The word *success* simply means the outcome being counted. A defective component may be called a success if X counts defects.

For exactly k successes,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Read the mathematics

Read this as: "The probability of exactly k successes equals the number of ways to place those successes, times the probability of one such arrangement." The combination counts the arrangements; the powers supply the probability of each arrangement.

6.4.1 Anchor example: defects in a sample of 20

Each item has a 2.5% probability of being defective. Twenty independent items are inspected. Let

$$X = \text{number of defective items among the 20.}$$

We want the probability of finding at least two defectives.

Why this method?

There are 20 fixed trials, each item is classified defective/not defective, $p = 0.025$ is treated as constant, and the item outcomes are assumed independent. This is a binomial model.

"At least two" means 2, 3, 4 and so on. It is shorter to exclude only 0 and 1:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

No defects

$$P(X = 0) = \binom{20}{0}(0.025)^0(0.975)^{20} = (0.975)^{20}.$$

DM42 keystrokes for $P(X = 0)$

[.975] [ENTER] [20] [y^x] [STO] [0] [0]
Display: **0.60268768**

Exactly one defect

$$P(X = 1) = \binom{20}{1}(0.025)(0.975)^{19}.$$

DM42 keystrokes for $P(X = 1)$

Combination: [20] [ENTER] [1] [f] [×] [COMB]
Multiply by p: [.025] [×]
Multiply by $(1 - p)^{19}$: [.975] [ENTER] [19] [y^x] [×]
Store the result: [STO] [0] [1]
Display: **0.30907061**

Take the complement

DM42 keystrokes

[1] [RCL] [0] [0] [-] [RCL] [0] [1] [-]
Display: **0.08824171**

Interpretation

Under the model, about 8.82% of comparable samples of 20 will contain two or more defective items. That is about nine samples in every hundred, not 8.82 defective items.

Sanity check

The expected number of defects is $np = 20(0.025) = 0.5$. Zero and one should therefore be common, while two or more should form a modest upper tail. The result is consistent with that picture.

6.4.2 Variation: communications availability

Each packet succeeds independently with probability 0.98. What is the probability of at least nine successes in ten attempts?

$$P(X \geq 9) = P(X = 9) + P(X = 10).$$

For nine successes,

$$P(X = 9) = \binom{10}{9}(0.98)^9(0.02) = 0.16674955.$$

DM42 keystrokes

10 ENTER 9 f × COMB .98 ENTER 9 y^x × .02 × STO 0 0

For ten successes,

DM42 keystrokes

.98 ENTER 10 y^x RCL 0 0 +
Display: **0.98382236**

Thus about 98.38% of ten-packet groups contain at least nine successful packets.

CHAPTER 7

Sampling without replacement: the hypergeometric model

The binomial model assumes independent trials. Drawing items from a finite lot without replacement breaks that assumption: after a defective item is drawn, both the lot composition and the next-draw probability change.

If a lot of N items contains K defectives and a sample of n is drawn, then

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Read the mathematics

The numerator counts samples containing exactly k defectives and $n - k$ good items. The denominator counts every possible sample of size n.

7.1 Anchor example: at least one defective

A lot contains 50 items, 4 of which are defective. Five items are selected without replacement. What is the probability that the sample contains at least one defective?

It is shorter to calculate the complement: no defectives means all five items come from the 46 good items.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{\binom{46}{5}}{\binom{50}{5}}.$$

DM42 keystrokes

Denominator: 50 ENTER 5 f × COMB STO 0 0
Numerator: 46 ENTER 5 f × COMB
Divide and complement: RCL 0 0 ÷ 1 $x \leftrightarrow y$ -
Display: **0.35303951**

Interpretation

There is about a 35.30% chance that the five-item sample finds at least one of the four defectives.

Variation– exactly two defectives

$$P(X=2)=\frac{\binom{4}{2}\binom{46}{3}}{\binom{50}{5}}=0.04298741.$$

DM42 keystrokes

4 ENTER 2 f × COMB 46 ENTER 3 f × COMB × RCL 0 0 ÷ →
0.04298741

CHAPTER 8

Poisson counts: events during an interval

The Poisson model describes a count of separate events occurring during a fixed interval of time, distance, area or volume. It is appropriate when the events are approximately independent and the average rate is reasonably stable throughout the interval.

Examples include calls per minute, flaws per metre of cable, particles detected per second and alarms per hour. It is less convincing when events arrive in clusters, the rate changes sharply, or one event makes another more likely.

If X is the count and λ is the expected count in the interval,

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

Read the mathematics

Read this as: "The probability of exactly k events equals e to the negative lambda, times lambda to the power k, divided by k factorial." Lambda is the average count for the interval. It is not a probability.

8.1 Anchor example: alarms during one hour

A monitoring system produces an average of 2.4 alarms per hour. The operations team wants to know how often four or more alarms will occur during a single hour.

Define

$$X = \text{number of alarms during one hour}, \qquad \lambda = 2.4.$$

The requested probability is

$$P(X \geq 4).$$

Read this as "the probability that X is at least four." It includes 4, 5, 6 and every higher count. It is shorter to remove the four low counts from the full probability of 1:

$$P(X \geq 4) = 1 - [P(0) + P(1) + P(2) + P(3)].$$

8.1.1 Start with zero alarms

For $k = 0$,

$$P(0) = e^{-2.4} = 0.09071795.$$

DM42 keystrokes

2.4 +/− e^x STO 0 0
Display: **0.09071795**

Rather than re-enter the full formula for every k, use the recurrence

$$P(k+1) = P(k)\frac{\lambda}{k+1}.$$

This says: multiply the previous probability by 2.4, then divide by the new count.

8.1.2 Generate the next three terms

DM42 keystrokes

$P(1)$ from $P(0)$: RCL 0 0 2.4 × 1 ÷ STO 0 1 → 0.21772309
$P(2)$ from $P(1)$: RCL 0 1 2.4 × 2 ÷ STO 0 2 → 0.26126771
$P(3)$ from $P(2)$: RCL 0 2 2.4 × 3 ÷ STO 0 3 → 0.20901416

8.1.3 Add the excluded cases and subtract from 1

DM42 keystrokes

RCL 0 0 RCL 0 1 + RCL 0 2 + RCL 0 3 + 1 $x \leftrightarrow y$ -
Display: **0.22127709**

Interpretation

About 22.13% of comparable one-hour periods are expected to contain four or more alarms. Put another way, roughly one out of every 4.5 one-hour periods will meet or exceed four alarms over the long run. This does not imply a regular alarm burst every 4.5 hours.

Sanity check

The average is 2.4 alarms, so 0 through 3 should account for more than half the hours, while 4 or more should remain quite possible. The calculated split—77.87% below four and 22.13% at least four—is sensible.

Variation– change the interval

If the rate remains stable and the question concerns 30 minutes, the expected count is $\lambda = 2.4(0.5) = 1.2$. Lambda must match the interval being studied. Using 2.4 in a half-hour calculation would silently answer the wrong question.

CHAPTER 9

The normal model: measurements around a centre

The normal distribution describes a continuous, symmetric bell-shaped model. Its centre is the mean μ and its spread is the standard deviation σ.

A measurement is standardized with

$$z = \frac{x - \mu}{\sigma}.$$

Read the mathematics

Read this as "z equals x minus mu, divided by sigma." The z value says how many standard deviations the measurement lies above or below the mean.

9.1 Anchor example: shaft tolerance

A stable shaft process is modelled with

$$\mu = 20.001 \text{ mm}, \qquad \sigma = 0.012 \text{ mm}.$$

The tolerance limits are 19.970 mm and 20.030 mm.

Standardize the lower limit:

DM42 keystrokes

19.970 ENTER 20.001 - .012 ÷

Display: **-2.58333333**

Standardize the upper limit:

DM42 keystrokes

20.030 ENTER 20.001 - .012 ÷

Display: **2.41666667**

The desired model probability is the area under the standard normal curve between these two z values:

$$P(19.970 \le X \le 20.030) = \Phi(2.4167) - \Phi(-2.5833) \approx 0.98727576.$$

The supplied PHI program evaluates the two cumulative probabilities. Its complete listing and explanation are printed in Appendix D. After entering that listing once:

DM42 keystrokes

Lower cumulative probability: 2.58333333 +/− XEQ PHI ENTER STO 0 4
Upper cumulative probability and difference: 2.41666667 XEQ PHI ENTER RCL 0 4 -
Display: **0.98727576**

Interpretation

The model predicts that about 98.73% of produced shafts fall within tolerance. This is a model-based fraction, not a guarantee and not the same as the observed fraction in one sample.

The core statistics menu does not contain a normal cumulative-distribution command. The printed program makes this example self-contained; a standard normal table or an independent integration remains a useful check.

Optional refresher: the 68–95–99.7 guide

For a normal model, roughly 68% lies within one standard deviation of the mean, 95% within two and 99.7% within three. These are quick checks, not replacements for a precise tail calculation. The present limits are about 2.4 to 2.6 standard deviations from the mean, so an in-tolerance probability near 99% is plausible.

CHAPTER 10

Random numbers and simulation

Simulation imitates a random process many times and observes how often the event of interest occurs. It is especially useful for checking intuition and for systems that are awkward to calculate exactly.

10.1 Uniform random numbers

f × RAN returns a pseudorandom value U satisfying

$$0 \le U < 1.$$

A nonzero seed entered with f × SEED produces a repeatable sequence. Seed 0 requests a time-based starting state.

10.2 Anchor example: a simulated die

Divide the interval from 0 to 1 into six equal pieces. The expression

$$\lfloor 6U \rfloor + 1$$

maps those pieces to the integers 1 through 6. The floor brackets mean "take the integer part."

Listing 10.1: DIE: return one simulated die result

```
LBL "DIE"
RAN
6
x
IP
1
+
RTN
```

Enter the listing in program mode. Run it by pressing XEQ, selecting `DIE`, and pressing ENTER. Repeat the call to generate additional rolls.

Sanity check

Every returned value must be an integer from 1 through 6. Over a few rolls the counts may be uneven. Over many rolls, each face should approach one-sixth of the total.

10.3 Variation: estimating π

Generate random points (x, y) in the unit square. A point lies inside the quarter-circle when

$$x^2 + y^2 \leq 1.$$

If m of N points lie inside, then

$$\hat{\pi} = 4\frac{m}{N}.$$

The supplied `MCPI` program implements this experiment. Its result varies with the random sequence. Increasing N generally stabilizes the estimate, but error shrinks only in proportion to $1/\sqrt{N}$: about 100 times as many trials are needed for about 10 times the stability.

What to carry forward

Simulation produces evidence, not certainty. Always compare a simulation with an exact answer when one is available. A disagreement is a valuable clue that the model, program or interpretation needs attention.

Part IV

Measurements and Descriptive Statistics

CHAPTER 11

Samples, data entry and what a summary can say

A *population* is the full collection we want to understand: all shafts from a process this week, all customers using a service, or every reading the sensor could produce under specified conditions. A *sample* is the subset actually observed.

Statistics summarise the sample. We often use them to learn about the population, but that step requires judgement about how the sample was obtained.

11.1 Parameters and statistics

Population quantities are called *parameters*; sample quantities are called *statistics*.

Quantity	Population	Sample
Mean	μ (mu)	$\bar{x}$ (x-bar)
Standard deviation	σ (sigma)	s
Proportion	p	$\hat{p}$ (p-hat)

The symbols help distinguish a fixed but often unknown population property from the sample value used to estimate it.

11.2 Accuracy and precision are different

Accuracy concerns closeness to the correct or accepted value. Precision concerns closeness of repeated measurements to one another.

Five readings from instrument A are 10.18, 10.19, 10.20, 10.21 and 10.22. Their mean is 10.200 and sample standard deviation is 0.01581. The readings are tightly grouped but high if the accepted value is 10.000: precise, but biased.

Instrument B gives 9.84, 9.93, 10.01, 10.08 and 10.14. Its mean is exactly 10.000, but $s = 0.11895$. It is accurate on average in this small sample but much less precise.

Interpretation

No single number captures both accuracy and precision. Report a measure of centre, a measure of spread and the reference or target value.

11.3 Representativeness comes before arithmetic

A perfectly calculated mean from a biased sample remains a biased description. Before summarising, ask:

- Were observations selected in a way that represents the population?
- Are repeated readings independent, or do they share drift or environment?
- Were units and instrument settings consistent?
- Were missing values or rejected readings handled transparently?

11.4 Entering statistical data safely

The DM42 accumulates statistical sums in ordinary numbered registers. The default compact layout occupies R11–R16; the extended layout uses R11–R23.

11.4.1 Clear the previous dataset

Before starting a new dataset:

DM42 keystrokes

f ← CLΣ

This clears the statistical accumulators without clearing unrelated registers.

11.4.2 One-variable data

Enter each measurement in X and press Σ+. For the values 4, 7 and 9:

DM42 keystrokes

4 Σ+ 7 Σ+ 9 Σ+

The small count returned after each entry is the number of accumulated observations.

To remove a value entered by mistake, enter the same value and use the shifted form Σ−:

DM42 keystrokes

7 f Σ+

This removes one occurrence of 7. Re-enter the correct value with Σ+.

11.4.3 Paired data

For a pair (x, y), the required order is

DM42 keystrokes

y ENTER x Σ+

The predictor x must be in X and the response y in Y when $\Sigma+$ is pressed. This order will matter in regression.

11.4.4 Choose the accumulator layout

DM42 keystrokes

Extended layout: f STAT ↓ ALLΣ
Compact layout: f STAT ↓ LINΣ

Use ALLΣ when logarithmic, exponential or power regression may be required. Do not store unrelated program data in the occupied register block.

CHAPTER 12

Mean and sample standard deviation

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Read the mathematics

Read this as "x-bar equals the sum of the observations divided by the number of observations." It is the balance point of the data.

The sample standard deviation is

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}.$$

It describes typical distance from the mean. The squared distances prevent positive and negative deviations from cancelling. The calculator's SDEV uses $n-1$, the sample definition.

12.1 Anchor example: resistor measurements

Twelve measured resistances, in kΩ, are:

9.98	10.01	10.03	9.99	10.02	10.00
9.97	10.04	10.01	9.99	10.02	10.00

First clear the statistical registers, then enter the values:

DM42 keystrokes

f ← CLΣ
9.98 Σ+ 10.01 Σ+ 10.03 Σ+ 9.99 Σ+
Continue in the printed order through the final 10.00.

Retrieve the mean:

DM42 keystrokes

f STAT MEAN
Display: $\bar{x} = 10.005000$ **kΩ in X**

Retrieve the sample standard deviation:

DM42 keystrokes

f STAT SDEV
Display: $s_x = 0.0206706$ **kΩ in X**

Retrieve the sum if desired:

DM42 keystrokes

f STAT SUM
Display: $\Sigma x = 120.06$ **kΩ in X**

Interpretation

The average measured resistance is 10.005 kΩ. The sample standard deviation is about 0.0207 kΩ, or 20.7 Ω. This describes the spread of these measurements; it does not by itself establish tolerance compliance or measurement accuracy.

Sanity check

The smallest reading is 9.97 and the largest is 10.04. A mean near 10.00 and a standard deviation much smaller than the 0.07 range are reasonable.

12.2 What MEAN, SDEV and SUM leave on the stack

For paired data, MEAN returns $\bar{x}$ in X and $\bar{y}$ in Y. SDEV returns s_x in X and s_y in Y. SUM returns Σx in X and Σy in Y. With one-variable data, use the X result.

CHAPTER 13

Weighted means: when observations represent unequal amounts

An ordinary mean gives every entered value the same influence. A weighted mean uses

$$\bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i},$$

where w_i is the weight attached to x_i.

Read the mathematics

Read this as "the weighted mean equals the sum of weight times value, divided by the sum of the weights." A value representing 15 pieces should influence a production summary more than one representing 7 pieces.

13.1 Anchor example: three production shifts

Three shifts report the following mean resistance and number tested:

Shift	Mean (kΩ)	Pieces
Morning	9.98	8
Afternoon	10.02	15
Night	10.01	7

The correct combined mean is

$$\bar{x}_w = \frac{8(9.98) + 15(10.02) + 7(10.01)}{8 + 15 + 7} = 10.0070 \text{ kΩ}.$$

For WMEAN, enter weight in Y and value in X:

DM42 keystrokes

f ← CLΣ
8 ENTER 9.98 Σ+
15 ENTER 10.02 Σ+
7 ENTER 10.01 Σ+
f STAT WMEAN
Display: **10.0070 kΩ**

An unweighted mean of the three shift means is 10.0033 kΩ. It answers a different question: the mean of three shifts rather than the mean of 30 pieces.

CHAPTER 14

Outliers: investigate before excluding

An outlier is an observation noticeably separated from the rest. It may be a transcription error, an instrument problem, a rare but genuine event, or evidence that the process changed. The arithmetic alone cannot decide which.

14.1 Anchor example: shaft diameters

Fifteen diameters, in millimetres, are:

20.004	19.998	20.006	19.993	20.011
20.001	19.996	20.008	19.989	20.003
19.997	20.005	20.000	19.994	20.041

Entering all 15 values gives

$$\bar{x} = 20.003067 \text{ mm}, \qquad s = 0.012092 \text{ mm}.$$

Removing the final 20.041 mm observation gives

$$\bar{x} = 20.000357 \text{ mm}, \qquad s = 0.006234 \text{ mm}.$$

DM42 keystrokes

After calculating the full summary, remove the suspect value with 20.041 f Σ+
Then repeat f STAT MEAN and f STAT SDEV.

Interpretation

The possible outlier changes the mean by only 0.00271 mm but nearly doubles the estimated spread. Decisions based on variability are therefore highly sensitive to how this observation is handled.

14.2 A defensible outlier conversation

Ask these questions before exclusion:

1. Is the value impossible or merely unusual?
2. Does the original record confirm the decimal point and unit?
3. Was the instrument zeroed and within range?
4. Did the process, operator, material or environment change?
5. Was an exclusion rule specified before seeing the result?

If the cause remains uncertain, show both analyses and explain why.

CHAPTER 15

From a sample to a population

A sample statistic varies from one random sample to another. This is sampling variability. Larger representative samples generally give more stable estimates, but size cannot repair systematic bias.

15.1 A sample proportion

Suppose 14 of 200 inspected units fail. The sample failure proportion is

$$\hat{p} = \frac{14}{200} = 0.07.$$

DM42 keystrokes

14 ENTER 200 ÷ → 0.07

A rough standard error for a proportion is

$$\mathrm{SE}(\hat{p}) \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Read the mathematics

Read this as "the standard error of p-hat is approximately the square root of p-hat times one minus p-hat, divided by n." It estimates how much the sample proportion would vary across repeated random samples of the same size.

For $\hat{p} = 0.07$ and $n = 200$:

DM42 keystrokes

.07 ENTER 1 $x \leftrightarrow y$ - .07 × 200 ÷ $\sqrt{x}$
Display: **0.0180416**

The estimated sampling uncertainty is about 0.018, or 1.8 percentage points. This does not include bias from an unrepresentative inspection process.

What to carry forward

Descriptive statistics say what the sample looks like. Inference asks what the sample supports about a wider population. Keep those two claims separate in reports.

Part V

Paired Data and Regression

CHAPTER 16

Understanding and fitting a straight line

Paired data record two quantities on the same object, time or trial. Examples include input voltage and sensor output, distance and delivery time, or pendulum length and period. Pairing is part of the information: an x value must remain attached to its corresponding y value.

16.1 Predictor and response

We usually call the explanatory or predictor quantity x and the response quantity y. This naming expresses the question, not necessarily a causal law.

In sensor calibration:

$$x = \text{reference input voltage}, \qquad y = \text{sensor output in mV}.$$

The aim is to describe or predict output from a known input.

16.2 A scatter plot comes before a fit

A scatter plot places each pair (x_i, y_i) as a point. Before calculating a correlation or regression, look for:

- direction: does y generally rise or fall as x rises?
- form: straight, curved, grouped or irregular?
- spread: is variability roughly constant?
- unusual points: is one pair separated from the rest?
- range: where were observations actually made?

16.3 Correlation

The correlation coefficient r measures the direction and strength of a *linear* association. It lies between -1 and 1.

Value	Interpretation
r near 1	strong positive linear association
r near -1	strong negative linear association
r near 0	little linear association; a curved relation may still exist

Correlation is unitless. It does not tell us the slope, and it does not prove causation. A shared cause, selection effect or time trend can produce correlation between quantities that do not directly affect one another.

16.4 Enter pairs in the correct orientation

For every pair, enter y first, then x:

DM42 keystrokes

[y] [ENTER] [x] [$\Sigma+$]

The calculator accumulates the pair with x in X and y in Y. If the axes are reversed, the fitted prediction question also reverses.

16.5 Linear regression: calibrating a sensor

The fitted straight-line model is

$$\hat{y} = mx + b.$$

Read the mathematics

Read this as "y-hat equals m x plus b." The hat marks a fitted or predicted response. The slope m is the predicted change in y for one unit of x; the intercept b is the fitted value when $x = 0$.

16.5.1 Anchor example: voltage calibration

The following reference inputs and sensor outputs were recorded:

x (V)	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y (mV)	251	503	752	1006	1253	1505	1757	2004

x (V)	y (mV)
4.5	2259

Clear and enter the pairs

DM42 keystrokes

[f] [←] [CLΣ]
First pair: [251] [ENTER] [.5] [$\Sigma+$]
Second pair: [503] [ENTER] [1] [$\Sigma+$]
Third pair: [752] [ENTER] [1.5] [$\Sigma+$]
Continue in table order through [2259] [ENTER] [4.5] [$\Sigma+$].

Select the straight-line model

DM42 keystrokes

[f] [STAT] [CFIT] [MODL] [LINF]

Read the fitted coefficients

DM42 keystrokes

Slope: f STAT CFIT SLOPE $\rightarrow m = 501.4666667$
Intercept: f STAT CFIT YINT $\rightarrow b = 0.7777778$
Correlation: f STAT CFIT CORR $\rightarrow r = 0.999997355$

The fitted equation is

$$\hat{y} = 501.466667x + 0.777778 \quad \text{mV}.$$

Interpretation

For each additional volt of reference input, the fitted sensor output rises by about 501.47 mV. The intercept is about 0.78 mV. The correlation indicates an extremely strong linear relationship over the measured range of 0.5 to 4.5 V.

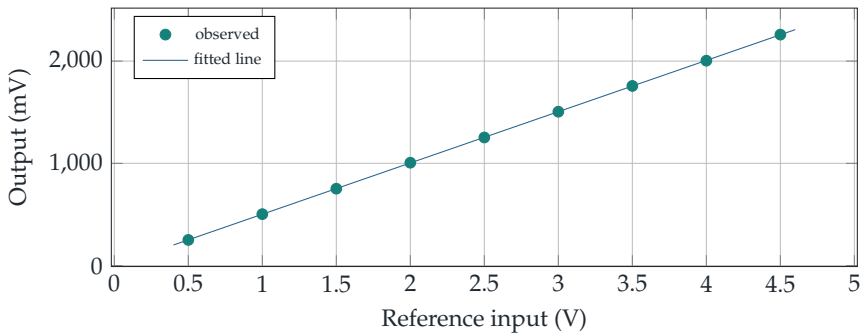


Figure 16.1: Calibration observations and fitted line.

16.5.2 Forecast output from a known input

For an input of 2.75 V:

DM42 keystrokes

2.75 f STAT CFIT FCSTY
Display: **1379.8111 mV**

This is interpolation: 2.75 V lies inside the observed 0.5–4.5 V range.

16.5.3 Estimate input from an observed output

For an output of 1800 mV:

DM42 keystrokes

1800 f STAT CFIT FCSTX
Display: **3.5879199 V**

Sanity check

A slope near 500 mV/V means 1800 mV should correspond to a little under 3.6 V. Both forecasts agree with the scale visible in the table and graph.

16.6 Residuals: what the line failed to explain

For observation i, the residual is

$$e_i = y_i - \hat{y}_i.$$

Read the mathematics

Read this as "residual i equals observed y minus predicted y." A positive residual means the observation lies above the fitted line; a negative residual lies below it.

16.6.1 Calculate one residual

At $x = 2.0$ V, the observed output is 1006 mV. The line predicts 1003.7111 mV.

DM42 keystrokes

Prediction: 2 f STAT CFIT FCSTY
Observed minus predicted: 1006 $x \leftrightarrow y$ -
Display: **2.2888889 mV**

The complete residual pattern is:

x (V)	Observed y	Predicted $\hat{y}$	Residual e
0.5	251	251.511	−0.511
1.0	503	502.244	0.756
1.5	752	752.978	−0.978
2.0	1006	1003.711	2.289
2.5	1253	1254.444	−1.444
3.0	1505	1505.178	−0.178
3.5	1757	1755.911	1.089
4.0	2004	2006.644	−2.644
4.5	2259	2257.378	1.622

The residuals alternate around zero and remain small relative to the roughly 2,000 mV response span. No obvious curvature or widening appears in this small dataset.

16.6.2 Patterns that deserve attention

Curved residual pattern
A straight line is missing systematic curvature.

Funnel shape
The response variability changes with x.

Long run of one sign
Drift, time order or a missing variable may be present.

One very large residual
Check the observation, experimental conditions and model assumptions.

16.6.3 Interpolation and extrapolation

Interpolation predicts within the observed range. Extrapolation predicts outside it. The fitted equation will produce a number at 9 V, but the data do not establish that the sensor remains linear there. Always state the measured range beside an extrapolated result.

CHAPTER 17

Exponential regression: a discharging capacitor

A capacitor discharge follows the physical form

$$V(t) = V_0 e^{-t/\tau}.$$

The DM42 exponential model is

$$y = be^{mx}.$$

Comparing the two forms gives $b \approx V_0$ and $m = -1/\tau$.

17.1 Anchor example

Time t (s)	0	1	2	3	4	5	6
Voltage V (V)	5.02	3.64	2.67	1.95	1.41	1.04	0.75

Enter every pair as voltage ENTER time Σ+. Then select EXPF:

DM42 keystrokes

f STAT CFIT MODL EXPF
SLOPE → $m = -0.315976812$
YINT → $b = 5.01423269$
CORR → $r = -0.999978282$

The fitted model is

$$\hat{V} = 5.01423269e^{-0.315976812t}.$$

Because $m = -1/\tau$,

$$\tau = -\frac{1}{m} = 3.1647892 \text{ s}.$$

DM42 keystrokes

After retrieving the slope: 1/x +/−
Display: **3.1647892 s**

At 2.5 s:

DM42 keystrokes

2.5 f STAT CFIT FCSTY
Display: **2.2758153 V**

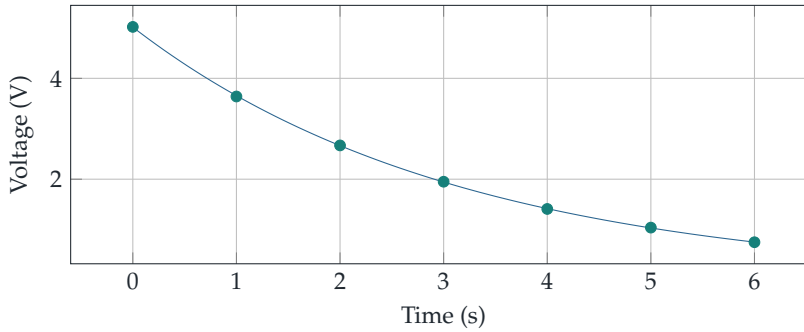


Figure 17.1: Measured discharge and fitted exponential curve.

Interpretation

The fitted initial voltage is about 5.014 V and the time constant about 3.165 s. At one time constant, an ideal exponential discharge has fallen to roughly 36.8% of its initial value.

CHAPTER 18

Power and logarithmic regression

Nonlinear models should be chosen because their form makes sense for the system, not merely because one correlation coefficient is slightly larger.

18.1 Power law: pendulum period

A simple pendulum predicts

$$T = 2\pi\sqrt{\frac{L}{g}} = bL^m,$$

with theoretical exponent $m = 1/2$.

L (m)	.20	.30	.40	.50	.70	.90	1.10	1.30
T (s)	.897	1.100	1.269	1.421	1.679	1.900	2.102	2.288

After entering the pairs:

DM42 keystrokes

f STAT CFIT MODL PWRF
SLOPE → $m = 0.499375064$
YINT → $b = 2.00561250$
CORR → $r = 0.999995397$

The exponent closely matches the theoretical $1/2$. From $b = 2\pi/\sqrt{g}$,

$$g = \frac{4\pi^2}{b^2} = 9.8144435\ \text{m/s}^2.$$

18.2 Logarithmic law: sound level and power

Sound level in decibels changes logarithmically with power. Measured pairs are:

Power (W)	.1	.2	.5	1	2	5	10	20
Level (dB)	69.8	72.9	76.7	79.8	82.9	86.6	89.7	92.8

The LOGF model is

$$y = m\ln x + b.$$

DM42 keystrokes

f STAT CFIT MODL LOGF
SLOPE → $m = 4.32230522$
YINT → $b = 79.7814413$
CORR → $r = 0.999941008$

Thus

$$\widehat{\text{level}} = 4.32230522 \ln(\text{power}) + 79.7814413.$$

18.3 Domain restrictions

Model	Form	Required positive values
LOGF	$y = m \ln x + b$	$x > 0$
EXPF	$y = be^{mx}$	$y > 0$
PWRF	$y = bx^m$	$x > 0$ and $y > 0$

These restrictions come from the logarithmic transformations used in fitting. A zero or negative observation is not an inconvenience to discard; it may show that the proposed model is inappropriate.

18.4 Why BEST is only a starting suggestion

The BEST option compares absolute correlations after the candidate transformations. Over a narrow range, several models may fit almost equally well. In the six-point example printed in Appendix F, the linear correlation is 0.999288 and the exponential transformed correlation is 0.999417. The numerical difference is tiny. Mechanism, residuals, domain and extrapolation behaviour should decide.

What to carry forward

Use correlation to describe linear agreement in the coordinates used for the fit. Use physical reasoning and residuals to decide whether the model deserves to be trusted.

Part VI

Using the Results to Decide

CHAPTER 19

Reliability structures

Reliability is a probability: the chance that a system performs its required function for a stated time under stated conditions. The system structure determines how component probabilities combine.

19.1 Series: every component is required

For independent components in series,

$$R_{\text{series}} = \prod_i R_i.$$

The product symbol means multiply all component reliabilities.

For reliabilities 0.98, 0.97 and 0.995:

DM42 keystrokes

[.98] [ENTER] [.97] [×] [.995] [×]
Display: **0.945847**

One necessary component failing stops the series system.

19.2 Parallel: at least one path is required

For independent parallel components,

$$R_{\text{parallel}} = 1 - \prod_i (1 - R_i).$$

This is the complement of every parallel path failing.

Two units with reliability 0.92 give

$$R = 1 - (0.08)^2 = 0.9936.$$

DM42 keystrokes

[1] [ENTER] [.92] [-] [x^2] [1] [$x \leftrightarrow y$] [-]
Display: **0.9936**

19.3 Anchor example: a 3-of-4 voting system

Four independent channels each work with probability $p = 0.9$. The system works if at least three channels work.

Let X be the number of working channels. Then

$$P(X \geq 3) = P(X = 3) + P(X = 4).$$

DM42 keystrokes

Exactly three: 4 ENTER 3 f × COMB .9 ENTER 3 y^x × .1 × STO 0 0
All four: .9 ENTER 4 y^x RCL 0 0 +
Display: **0.9477**

Interpretation

Under independent failures, the voting arrangement raises reliability from 90% per channel to 94.77% for the system.

Take care

Redundancy may not protect against common-cause failures such as shared power, shared software, fire, flooding or an incorrect maintenance procedure. Independence must be defended from the design, not assumed because the multiplication is convenient.

CHAPTER 20

Inventory as an expected-cost decision

A decision under uncertainty requires three ingredients: possible future states, their probabilities and the cost of each decision-state combination.

20.1 Anchor example: how many spare units?

Demand D for tomorrow can be 0 through 5 units:

D	0	1	2	3	4	5
$P(D)$	.05	.15	.30	.28	.15	.07

An unsold unit costs R80 to carry. An unmet unit of demand costs R220. For order quantity q,

$$C(q, D) = 80 \max(q - D, 0) + 220 \max(D - q, 0).$$

Read the mathematics

If stock exceeds demand, the first term charges for leftovers. If demand exceeds stock, the second term charges for shortage. Only one term is positive for any demand value.

Consider $q = 3$. The costs and probability-weighted costs are:

D	Probability	Cost	Probability × cost
0	.05	R240	R12.00
1	.15	R160	R24.00
2	.30	R80	R24.00
3	.28	R0	R0.00
4	.15	R220	R33.00
5	.07	R440	R30.80
		Expected cost	R123.80

DM42 keystrokes

Accumulate the weighted costs directly:

[.05] [ENTER] [240] [×] [.15] [ENTER] [160] [×] [+] [.30] [ENTER] [80] [×] [+] [.15] [ENTER] [220] [×] [+] [.07] [ENTER] [440] [×] [+]

Display: **123.80**

Repeating the calculation for $q = 0$ through 5 gives:

Order q	Expected cost
0	R558.80
1	R353.80
2	R193.80
3	R123.80
4	R137.80
5	R196.80

The smallest expected cost occurs at $q = 3$. This recommendation depends on the stated shortage and holding costs. If shortage becomes more expensive, the preferred order quantity can rise.

CHAPTER 21

Prediction in operations: delivery time

Ten observed trips pair distance with delivery time:

km	4	7	9	12	15	18	22	26	30	35
min	18	22	24	29	34	37	43	50	55	63

Enter each pair as minutes ENTER kilometres $\Sigma+$. After the last pair, select the straight-line model and retrieve its three summaries:

DM42 keystrokes

f STAT CFIT MODL LINF
f STAT CFIT SLOPE → 1.45667643 min/km
f STAT CFIT YINT → 11.5711595 min
f STAT CFIT CORR → 0.999308699

The fitted equation is

$$\widehat{\text{minutes}} = 1.45667643(\text{km}) + 11.5711595.$$

At 20 km:

DM42 keystrokes

20 f STAT CFIT FCSTY
Display: **40.704688 min**

Interpretation

Within this operating context, the fit predicts about 40.7 minutes for a 20 km trip. The slope describes additional travel time per kilometre. The intercept loosely represents fixed handling and route overhead, but it should not be treated as a universal constant.

Distance is not the only influence. Traffic, route class, loading, weather and stop count may all matter. A high correlation does not make those variables disappear.

CHAPTER 22

Quality decisions: observed versus modelled yield

Tolerance limits belong to the product specification. Standard deviation belongs to the process or measurement distribution. They are related in a capability calculation, but they are not interchangeable.

22.1 Observed yield

If 197 of 200 inspected items are within tolerance, the observed sample yield is

$$\hat{p} = \frac{197}{200} = 0.985 = 98.5\%.$$

DM42 keystrokes

197 ENTER 200 ÷
Display: **0.985, or 98.5%**

This is a direct description of that sample.

22.2 Modelled yield

If the process is modelled as normal with $\mu = 20.001$ mm and $\sigma = 0.012$ mm, the modelled probability between 19.970 and 20.030 mm is 98.7276%. This is an area under a fitted distribution.

The two values answer different questions:

- 98.5%: what fraction of the inspected sample passed?
- 98.7276%: what fraction does the normal model predict over repeated production?

22.3 Capability indices as compact summaries

When a process is approximately normal and stable,

$$C_p = \frac{USL - LSL}{6\sigma}$$

compares tolerance width with process spread, while

$$C_{pk} = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right)$$

also accounts for off-centre operation.

For the shaft example:

DM42 keystrokes

C_p: 20.030 ENTER 19.970 - 6 ENTER .012 × ÷ → 0.8333333

For the upper side,

DM42 keystrokes

20.030 ENTER 20.001 - 3 ENTER .012 × ÷ → 0.8055556

For the lower side,

DM42 keystrokes

20.001 ENTER 19.970 - 3 ENTER .012 × ÷ → 0.8611111

Therefore $C_{pk} = 0.8056$.

Interpretation

Both indices are below 1, indicating that the six-standard-deviation process width exceeds the tolerance width and that the upper side is slightly more restrictive. Capability indices are meaningful only after stability and distribution assumptions have been examined.

22.4 A complete field workflow

The following sequence is designed for a small engineering or operational dataset.

22.4.1 1. State the practical decision

Write the question, response, predictor, units, interval and decision threshold. "Analyse the data" is not yet a decision question.

22.4.2 2. Preserve and inspect the observations

Check units, decimal points, missing records, time order and pairing. Make a simple plot when two variables are involved.

22.4.3 3. Clear and enter deliberately

Use f ← CLΣ. For pairs, say the order aloud: "response ENTER predictor Σ+." Confirm the count after the final entry.

22.4.4 4. Summarise before modelling

Record n, mean, sample standard deviation, minimum, maximum and any unusual observations. A model should not be the first time you look at the data.

22.4.5 5. Choose from the mechanism

Use a straight line for approximately constant additive change, an exponential for constant percentage change, a power law for scale relationships and a logarithmic form for diminishing additive response to multiplicative input.

22.4.6 6. Calculate and record the complete result

Keep units with coefficients. Record the equation, correlation, data range and prediction. Do not report only r.

22.4.7 7. Examine residuals and assumptions

Look for curvature, changing spread, drift and influential observations. State independence, normality or constant-rate assumptions when they matter.

22.4.8 8. Interpret in ordinary language

Answer the practical question. A probability should identify the event and interval; a forecast should identify its units and range; an expected cost should identify the portfolio or time horizon.

22.4.9 9. Perform an independent check

Use a complement, direct multiplication, a rough scale estimate, a graph or a second calculation. A check should use a different route from the main calculation whenever possible.

22.4.10 10. Preserve the trail

Keep the raw observations, exclusions, settings, assumptions and rounded reported result. A future reader should be able to reproduce both the arithmetic and the reasoning.

CHAPTER A

Symbol and language glossary

This glossary is arranged for reading rather than memorising. The middle column gives a natural way to say an expression aloud; the last column states what it means in this book. A symbol can acquire a more specialised meaning in another field, so always read it with its surrounding sentence.

A.1 Sets, events and probability

Symbol	Read it as	Meaning here
A, B	"event A", "event B"	Named outcomes or collections of outcomes.
$P(A)$	"the probability of A"	A number from 0 to 1 measuring how likely event A is.
$A \cap B$	"A intersection B" or "A and B"	Both events occur. The upside-down U shape is the intersection sign.
$A \cup B$	"A union B" or "A or B"	At least one event occurs, including the possibility that both occur.
A^c	"A complement" or "not A"	Every outcome outside A. Thus $P(A^c) = 1 - P(A)$.
$P(A \mid B)$	"the probability of A given B"	The probability of A after restricting attention to cases in which B has occurred. The vertical bar means *given*, not division.
$A \subseteq B$	"A is a subset of B"	Every outcome in A also belongs to B.
$\varnothing$	"the empty set"	An impossible event containing no outcomes.
Ω	"omega" or "the sample space"	The complete set of outcomes under consideration.
$A \perp B$	"A is independent of B"	Knowing whether one occurred does not change the probability of the other.

Read the mathematics

The statement $P(A \cap B) = P(A)P(B)$ says: "The probability that A and B both occur equals the probability of A times the probability of B." It is valid when the events are independent. If they are not independent, use $P(A \cap B) = P(A)P(B \mid A)$.

A.2 Numbers, comparisons and ranges

Symbol	Read it as	Meaning here
$=$	“equals”	The two expressions have exactly the same mathematical value.
$\approx$	“is approximately equal to”	The displayed value is rounded or comes from an approximation.
$\neq$	“does not equal”	The values differ.
$<, >$	“less than”, “greater than”	A strict comparison; equality is excluded.
$\leq, \geq$	“less than or equal to”, “greater than or equal to”	A comparison that includes the boundary.
$a \leq X \leq b$	“X is between a and b, inclusive”	Both limits are included.
$X \in A$	“X is in A”	The value of X belongs to set A.
$\lvert x \rvert$	“absolute value of x”	The distance from zero; unlike a conditional-probability bar, two bars enclose a number.
$\max(a, b)$	“the maximum of a and b”	The larger of the two values.
$\min(a, b)$	“the minimum of a and b”	The smaller of the two values.
$\lfloor x \rfloor$	“floor of x”	The greatest integer not exceeding x.
∞	“infinity”	An unbounded direction or limit, not an ordinary number to enter.

A.3 Data and distribution symbols

Symbol	Read it as	Meaning here
X	“capital X”	A random variable: a numerical result not known in advance.
x	“lower-case x”	A particular observed value of X, or a predictor in regression.
x_i	“x sub i”	Observation number i; the subscript is a label, not multiplication.
n	“n”	Number of observations or number of trials.
k	“k”	A specified count of successes or selected objects.
p	“p”	A model probability or population proportion.
$\hat{p}$	“p hat”	A proportion calculated from a sample. The hat signals an estimate.
μ	“mu”	Population or model mean.
$\bar{x}$	“x bar”	Sample mean.
σ	“sigma”	Population or model standard deviation.
s	“s”	Sample standard deviation.
s^2	“s squared”	Sample variance.
λ	“lambda”	Mean event count in a Poisson interval.
z	“z” or “z-score”	Number of standard deviations an observation lies from the mean.
$E(X)$	“expected value of X”	The probability-weighted long-run average of X.
$\mathrm{Var}(X)$	“variance of X”	The expected squared distance from the mean.
SE	“standard error”	The modelled standard deviation of an estimator across repeated samples.
$P(X = k)$	“probability that X equals k”	Probability of one exact discrete value.
$P(X \geq k)$	“probability that X is at least k”	Probability of k or any larger value.

A.4 Operations and models

Symbol	Read it as	Meaning here
$\sum$	"sum"	Add a sequence of terms.
$\prod$	"product"	Multiply a sequence of terms.
$n!$	"n factorial"	The product $n(n-1)\cdots 2\cdot 1$, with $0! = 1$.
$\binom{n}{k}$	"n choose k"	Number of unordered selections of k objects from n.
e^x	"e to the x"	The natural exponential function.
$\ln x$	"natural log of x"	The power to which e must be raised to obtain x.
$\sqrt{x}$	"square root of x"	The non-negative number whose square is x.
$\hat{y}$	"y hat"	A value predicted by a fitted model.
m	"m" or "slope"	Change in predicted y for a one-unit increase in x in a straight-line model.
b	"b" or "intercept"	Predicted value of y when $x = 0$ in a straight-line model.
r	"r" or "correlation"	Strength and direction of a linear association, from -1 to 1.
R	"reliability"	Probability that a component or system performs its required function.

What to carry forward

Do not try to decode an entire formula in one glance. First name the symbols, then identify the operation, then say what event or quantity the whole expression describes.

A.5 Formula phrasebook

Each entry has three layers: the compact mathematical expression, an ordinary way to read it aloud, and the practical meaning. The third layer is the one to write in a report.

A.5.1 Probability

Expression	Read aloud	Interpretation
$P(A^c) = 1 - P(A)$	"Probability of not A is one minus probability of A."	Everything is either in A or outside it, so the two probabilities total 1.
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	"Probability of A or B is A plus B minus A and B."	The overlap was counted twice and must be subtracted once.
$P(A \mid B) = P(A \cap B)/P(B)$	"Probability of A given B is A and B divided by B."	Among the cases satisfying B, take the fraction that also satisfy A.
$P(A \cap B) = P(A)P(B \mid A)$	"A and B equals A times B given A."	Follow the first event by the conditional chance of the second.
$E(X) = \sum_x xP(X = x)$	"Expected X is the sum of each value times its probability."	A probability-weighted long-run average, not a promise for the next case.

A.5.2 Counting and distributions

Expression	Read aloud	Interpretation
$\binom{n}{k} = n!/[k!(n-k)!]$	"n choose k."	The number of ways to select k items when order does not matter.
$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	"For binomial X, the probability of k successes is n choose k times p to k times one minus p to n minus k."	Exactly k successes in n independent equal-probability trials.
$P(X = k) = e^{-\lambda}\lambda^k/k!$	"For Poisson X, probability of k is e to minus lambda times lambda to k over k factorial."	Exactly k events in an interval with mean count λ.
$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	"Choose k successes and n minus k failures, over all samples of n."	Exactly k successes in sampling without replacement.
$z = (x - \mu)/\sigma$	"z equals x minus mu, divided by sigma."	The signed distance from the mean measured in standard deviations.

A.5.3 Statistics and regression

Expression	Read aloud	Interpretation
$\bar{x} = \frac{1}{n}\sum_i x_i$	"x bar is the sum of the observations divided by n."	The sample's arithmetic centre.
$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$	"s is the square root of the sum of squared deviations over n minus one."	Typical sample spread around the sample mean.
$SE(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n}$	"standard error of p hat is the square root of p hat times one minus p hat over n."	Estimated sample-to-sample variability of a proportion under simple sampling.
$\hat{y} = mx + b$	"y hat equals m x plus b."	The response predicted by a fitted straight line.
$e_i = y_i - \hat{y}_i$	"residual i is observed y minus predicted y."	The vertical prediction error for observation i.
$\hat{y} = be^{mx}$	"y hat equals b times e to m x."	An exponential fit: equal increases in x multiply the response by a constant factor.
$\hat{y} = bx^m$	"y hat equals b times x to m."	A power fit: a percentage change in x produces a roughly constant percentage change in y.

A.5.4 Reliability and decisions

Expression	Read aloud	Interpretation
$R_s = \prod_i R_i$	"Series reliability is the product of the component reliabilities."	Every required independent component must work.
$R_p = 1 - \prod_i (1 - R_i)$	"Parallel reliability is one minus the product of the failure probabilities."	At least one independent parallel path must work.
$C_p = (USL - LSL)/(6\sigma)$	"C p is the specification width divided by six sigma."	Compares allowed width with modelled process spread.
$C_{pk} = \min\left(\frac{USL-\mu}{3\sigma}, \frac{\mu-LSL}{3\sigma}\right)$	"C p k is the smaller of the upper and lower capability ratios."	Reports the more restrictive specification side and includes off-centre operation.
$E[C(q,D)] = \sum_d C(q,d)P(D=d)$	"Expected cost is the sum of each cost times the probability of its demand."	Long-run average cost for decision q under the stated demand model.

CHAPTER B

DM42 keystroke atlas

This appendix is a compact route map. Menu names are shown in text; physical key presses use keycaps. Firmware variations may change a menu position, so read the visible menu label rather than counting positions by memory.

B.1 RPN essentials

Enter two numbers for a binary operation. Type the first number, press [ENTER], type the second, then press the operation. For $20 - 3$, use [20] [ENTER] [3] [-].

Reuse a displayed result. Type the next number and press the operation; the result already in the X-register becomes the first argument.

Swap an order. Use [$x \leftrightarrow y$] when the two top stack values need to change places.

Undo a mistaken digit. Use the normal correction key before committing the number. If a statistical observation has already been entered, remove that observation with the matching data order and [$\Sigma-$].

B.2 Probability and counting routes

Task	Route or pattern
Factorial	Enter n, then [f] [×] [$n!$].
Combinations	Enter n [ENTER] k, then [f] [×] [COMB].
Permutations	Enter n [ENTER] k, then [f] [×] [PERM].
Power a^b	Enter a [ENTER] b [y^x].
Natural exponential	Enter x, then [g] [e^x].
Natural logarithm	Enter a positive x, then [g] [LN].
Random number	Use the probability menu's RAN item.
Set random seed	Enter the seed and select SEED. Record it when reproducibility matters.

B.3 One-variable statistics

1. Clear old accumulations with [f] [←] [CLΣ].
2. For each observation, type its value and press [$\Sigma+$].
3. Open [f] [STAT]; select the one-variable statistics page.
4. Read n, $\bar{x}$, s, σ, sums and extrema from the labelled items.

Use sample s when observations are treated as a sample from a wider process. Use population σ only when the entered observations are the complete population of interest or when the problem explicitly asks for that divisor.

B.4 Paired statistics and curve fitting

For a pair with predictor x and response y, enter

$$[y]\ \text{ENTER}\ [x]\ [\Sigma+].$$

Repeat for every pair. The order is worth saying aloud: "response, ENTER, predictor, sigma plus."

Then use [f] STAT CFIT. To change the model, select MODL and then one of the first five items below. The result and forecast items are selected directly from CFIT.

Menu item	Use
LINF	Straight line $\hat{y} = mx + b$
LOGF	Logarithmic form $\hat{y} = m \ln x + b$
EXPF	Exponential form $\hat{y} = be^{mx}$
PWRF	Power form $\hat{y} = bx^m$
BEST	Numerically strongest listed correlation; inspect the mechanism too
SLOPE	Fitted slope or transformed slope
YINT	Fitted intercept or transformed intercept
CORR	Correlation for the selected transformed relation
FCSTY	Predict y from an entered x
FCSTX	Estimate x from an entered y

Take care

For EXPF and PWRF, the displayed coefficients describe the calculator's model form. Write the equation and test a prediction before reporting it. A strong correlation does not establish causation or justify extrapolation.

B.5 Running the supplied programs

After entering a labelled program, return to normal calculator mode. To run the die simulator, press XEQ, select or spell the label DIE, then press ENTER. To run the Monte Carlo program, enter the number of trials, press XEQ, select or spell MCPI, and press ENTER. The program listings and line-by-line explanations are in Appendix D.

B.6 The statistical registers and result order

The calculator keeps accumulated totals in statistical registers. You normally retrieve them through labelled statistics functions; the map below helps when checking a program or diagnosing an entry problem.

B.6.1 What one paired entry stores

Entering y [ENTER] x [Σ+] updates counts and sums equivalent to

$$n, \quad \sum x, \quad \sum x^2, \quad \sum y, \quad \sum y^2, \quad \sum xy.$$

From these totals the calculator can derive means, standard deviations, correlation and fitted coefficients without storing every original pair.

Read the mathematics

Read $\sum x^2$ as "the sum of the squared x values." It is not the same as $(\sum x)^2$, which is "the square of the sum." For $x = 2, 3$, the first is $2^2 + 3^2 = 13$ and the second is $(2+3)^2 = 25$.

B.6.2 Why entry order matters

The DM42 convention makes x the value in the X-register when [Σ+] is pressed and y the value above it in the Y-register. Thus response y is entered first, then predictor x. Reversing every pair fits the inverse relationship, which has different slope, intercept and forecast behaviour even though the magnitude of correlation is unchanged.

B.6.3 Useful result-order checks

Result	Check
Count n	Must equal the number of observations or pairs entered.
Mean	Must lie between the smallest and largest entered values.
Standard deviation	Cannot be negative; it is zero only when all values are identical.
Correlation r	Must lie between -1 and 1.
Combination $\binom{n}{k}$	Is an integer and is symmetric: $\binom{n}{k} = \binom{n}{n-k}$.
Probability	Must lie from 0 to 1 inclusive.
Forecast	Substitute it back into the displayed model when practical.

B.6.4 Clearing, correcting and preserving

Clear statistical accumulations before starting a new dataset. If the most recent entered observation is wrong, re-enter exactly the same value or pair in the same order and press [Σ−]; then enter the corrected item with [Σ+]. Preserve the original observations outside the calculator, because a register total cannot reconstruct their order.

B.7 Choosing a model without guessing

Start with the physical or operational mechanism, then check whether the data support it. A familiar formula is not automatically the right formula.

B.7.1 Count and probability models

Model	Use when	Questions to ask
Binomial	A fixed number of trials, two outcomes per trial, constant success probability, and independent trials.	Is n genuinely fixed? Does one result affect the next? Is p stable?
Hypergeometric	A fixed sample is drawn without replacement from a finite population with known success count.	Does removing one item change the next probability? Are N, K and n known?
Poisson	Events are counted over a stated interval with an approximately constant average rate and no clustering beyond the model.	What is the interval? Is the rate stable? Can simultaneous events be neglected?
Normal	A continuous variable is approximately symmetric and bell-shaped, or a justified sampling approximation applies.	Are tails and outliers plausible? Is the process stable? Is σ known or estimated?
Empirical proportion	The immediate question is the observed fraction in a sample.	Was the sample representative? How large is n? What uncertainty accompanies the proportion?

B.7.2 Regression models

Model	Shape and mechanism	Diagnostic thought
Linear $mx + b$	Approximately constant additive change in y per unit of x.	Do residuals wander randomly around zero rather than curve?
Exponential be^{mx}	Approximately constant percentage change for equal additive steps in x.	Does $\ln y$ versus x look linear, with positive y values?
Power bx^m	Scale relationship: percentage changes in x accompany approximately constant percentage changes in y.	Do $\ln y$ and $\ln x$ form a line, with positive values?
Logarithmic $m \ln x + b$	The response changes rapidly at small x and then by diminishing amounts.	Does y versus $\ln x$ look linear, with positive x?

Why this method?

The BEST item compares correlations after transformation. It cannot know whether a model respects units, causal direction, operating limits or the mechanism. Treat it as a prompt for inspection, not an automatic decision.

B.7.3 A five-question model check

1. What is random, and what is fixed by design?

2. What unit, population and time interval does the model describe?
3. Which independence, constant-rate or shape assumption is being made?
4. What observation would reveal that assumption to be poor?
5. Is the requested prediction inside the observed range?

CHAPTER C

Complete example data

All observations used in the worked statistical and regression examples are printed here. The tables are the authoritative, self-contained source for re-entering an example or checking an intermediate result.

C.1 One-variable data

C.1.1 Resistor measurements (kΩ)

Used for the mean and sample-standard-deviation example.

9.98	10.01	10.03	9.99	10.02	10.00
9.97	10.04	10.01	9.99	10.02	10.00

Count $n = 12$; sum 120.06 kΩ; mean 10.005 kΩ; sample standard deviation 0.0206706 kΩ.

C.1.2 Shaft diameters (mm)

Used for the outlier-sensitivity example. The last value is investigated as a possible unusual observation; it is not silently discarded.

20.004	19.998	20.006	19.993	20.011
20.001	19.996	20.008	19.989	20.003
19.997	20.005	20.000	19.994	20.041

With all 15 values, $\bar{x} = 20.0030667$ and $s = 0.0120917$ mm. Without the last value, $\bar{x} = 20.0003571$ and $s = 0.0062340$ mm.

C.2 Paired regression data

C.2.1 Sensor calibration

Predictor x is input voltage (V); response y is indicated units.

x	.5	1	1.5	2	2.5	3	3.5	4	4.5
y	251	503	752	1006	1253	1505	1757	2004	2259

Linear fit: $\hat{y} = 501.466667x + 0.777778$, $r = 0.999997355$.

C.2.2 RC discharge

Time t is in seconds and voltage V is in volts.

t	0	1	2	3	4	5	6
V	5.02	3.64	2.67	1.95	1.41	1.04	.75

Exponential fit: $\hat{V} = 5.01423269e^{-0.315976812t}$; estimated time constant $\tau = 3.16478920$ s.

C.2.3 Pendulum period

Length L is in metres and period T in seconds.

L	.20	.30	.40	.50	.70	.90	1.10	1.30
T	.897	1.100	1.269	1.421	1.679	1.900	2.102	2.288

Power fit: $\hat{T} = 2.00561250L^{0.499375064}$. The exponent is close to the square-root prediction from the small-angle model.

C.2.4 Sound power and level

Power is in arbitrary consistent units; level is in dB.

Power	.1	.2	.5	1	2	5	10	20
Level	69.8	72.9	76.7	79.8	82.9	86.6	89.7	92.8

Logarithmic fit: $\widehat{\text{level}} = 4.32230522\ln(\text{power}) + 79.7814413$.

C.2.5 Nearly linear model-selection trap

These values illustrate why a numerical ranking should not replace scientific judgement.

x	0	1	2	3	4	5
y	100.1	102.0	104.2	106.0	108.5	110.4

Over a short range, several smooth models may look almost indistinguishable.

C.2.6 Delivery distance and time

Distance is in kilometres; response is delivery time in minutes.

km	4	7	9	12	15	18	22	26	30	35
min	18	22	24	29	34	37	43	50	55	63

Linear fit: $\widehat{\text{minutes}} = 1.45667643(\text{km}) + 11.5711595$.

C.3 Decision tables

C.3.1 Inventory demand distribution

Demand	0	1	2	3	4	5
Probability	.05	.15	.30	.28	.15	.07

The probabilities total 1. The expected-cost example uses R80 per unsold unit and R220 per unmet unit.

CHAPTER D

Program listings and explanations

The programs are short enough to enter and audit line by line. Program steps are printed exactly in calculator-style notation; the prose beside them explains the purpose of each block.

D.1 DIE: simulate one fair die

```
LBL "DIE"
RAN
6
x
IP
1
+
RTN
```

RAN produces a uniform pseudo-random value from zero up to, but not including, one. Multiplying by 6 produces a value from zero up to, but not including, six. IP keeps its integer part, giving 0 through 5; adding 1 gives the required die result 1 through 6.

To run it after entry, return to normal mode and use XEQ DIE ENTER. Repeated runs form a sequence from the current random-number seed. For an auditable simulation, record and set the seed before the first run.

D.2 PHI: standard normal cumulative probability

The PHI program accepts a z value and returns an approximation to $\Phi(z) = P(Z \leq z)$. It uses a compact rational approximation whose error is small compared with the reported precision in the normal-distribution example.

Listing D.1: PHI: approximate the standard normal cumulative distribution

```
LBL "PHI"
STO 00
ABS
X^2
2
/
+/-
E^X
0.3989422804
x
STO 01
RCL 00
ABS
0.2316419
```

```
  x
  1
  +
  1/X
  STO 02
  1.330274429
  RCL 02
  x
  1.821255978
  +/-
  +
  RCL 02
  x
  1.781477937
  +
  RCL 02
  x
  0.356563782
  +/-
  +
  RCL 02
  x
  0.319381530
  +
  RCL 02
  x
  RCL 01
  x
  STO 03
  RCL 00
  0
  X>Y?
  GTO 01
  1
  RCL 03
  -
  RTN
  LBL 01
  RCL 03
  RTN
```

The program uses registers 00 through 03. Enter a z value, then use [XEQ] [PHI] [ENTER]. For the lower limit in the shaft example, $z = -2.58333333$ returns approximately 0.00489257; for the upper limit, $z = 2.41666667$ returns approximately 0.99216833. Their difference is 0.98727576.

D.3 MCPI: estimate π by random points

Enter a positive integer trial count before running the program.

```
  LBL "MCPI"
  STO 00
  0
  STO 01
  STO 02
  LBL 00
  RAN
  X^2
  RAN
  X^2
  +
  1
  X<>Y
  X<=Y?
  GTO 01
  GTO 02
  LBL 01
```

```
18  1
19  STO+ 01
20  LBL 02
21  1
22  STO+ 02
23  RCL 02
24  RCL 00
25  X>Y?
26  GTO 00
27  4
28  RCL 01
29  x
30  RCL 00
31  /
32  RTN
```

Register 00 holds the requested trial count, register 01 the number of points inside the unit quarter-circle, and register 02 the number of points processed. Each loop generates independent coordinates x, y in $[0, 1)$ and evaluates $x^2 + y^2$. A value no greater than 1 lies inside the quarter-circle. The area ratio estimates $\pi/4$, so the final line multiplies the inside fraction by 4.

To run 10,000 trials, use

10000 XEQ MCPI ENTER.

The answer will vary with the random sequence. Increasing the trial count usually improves accuracy, but Monte Carlo error decreases slowly: roughly in proportion to $1/\sqrt{n}$ rather than $1/n$.

Sanity check

The estimate should fall between 0 and 4 and ordinarily near 3.14 for a large run. A repeat with the same seed and trial count should reproduce the same result; a different seed tests sensitivity to the simulated sequence.

CHAPTER E

Field reference and reporting

Most unexpected results come from stale statistical accumulations, reversed pair order, a misunderstood event, or premature rounding. Work from the top of this list rather than re-entering everything blindly.

E.1 A result looks impossible

1. Check the range: probabilities belong in $[0, 1]$, counts are non-negative, and r belongs in $[-1, 1]$.
2. Re-read the event. "At least two" means 2, 3, and every larger possible count; "more than two" begins at 3.
3. Check units and scale. A percentage entered as 2.5 rather than 0.025 is a hundredfold error.
4. Recalculate by a different route. For "at least one", use the complement $1 - P(0)$.
5. Clear only after recording anything that must be preserved, then re-enter a small prefix and inspect the count.

E.2 The statistical count is wrong

If n is too large, an item may have been entered twice or old data were not cleared. If the incorrect item is known, remove it with $\boxed{\Sigma-}$. If the history is uncertain, preserve the source table, clear the statistical registers, and enter the data again while ticking off each row.

E.3 Regression coefficients look reversed

Confirm that each pair was entered as response y $\boxed{\text{ENTER}}$ predictor x $\boxed{\Sigma+}$. Confirm the selected fit type. Finally, substitute the first and last observed x values into the equation: predicted y values should be in the neighbourhood of the observed responses.

E.4 A displayed answer differs in the last digits

The display may be rounded even though internal registers retain more digits. Carry full calculator precision through intermediate work and round once, at the end. A final difference of one unit in the last reported decimal can be normal when two valid routes round intermediate quantities differently.

E.5 A model fits but makes no sense

A high $|r|$ says that the selected transformed relationship is close to a line over the observed range. It does not guarantee causation, correct units, independent residuals or safe extrapolation. Return to the mechanism and plot the residuals. Prefer a slightly weaker numerical fit that has a defensible physical meaning over an arbitrary curve selected only by ranking.

E.6 Pre-calculation checklist

- Event, population, interval and units are stated.
- Percentages have been converted to decimal probabilities where required.
- Old statistical accumulations have been cleared.
- Paired entry order is written at the top of the work.
- Assumptions are named, not merely implied.

E.7 Post-calculation checklist

- The answer lies in its allowable range and has units where appropriate.
- A complement, rough estimate, substitution or second method supports it.
- Rounding matches the decision rather than the maximum display length.
- The sentence reporting the result names the event or predicted quantity.
- Data range and important limitations accompany a forecast.

E.8 Formula reference

This appendix collects the formulas used in the book. Use the phrasebook and glossary when a symbol is unfamiliar; use the main chapter when an assumption or derivation is needed.

E.8.1 Counting and elementary probability

$$n! = n(n-1)\cdots 2\cdot 1, \qquad {}_nP_k = \frac{n!}{(n-k)!}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

$$P(A^c) = 1 - P(A),$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \cap B) = P(A)P(B \mid A), \qquad P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

For independent events, $P(A \cap B) = P(A)P(B)$.

E.8.2 Discrete distributions

Binomial:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad E(X) = np, \quad \operatorname{Var}(X) = np(1-p).$$

Hypergeometric:

$$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}},\quad E(X)=n\frac{K}{N}.$$

Poisson:

$$P(X=k)=e^{-\lambda}\frac{\lambda^k}{k!},\quad E(X)=\lambda,\quad \mathrm{Var}(X)=\lambda.$$

The Poisson recurrence is

$$P(X=k+1)=P(X=k)\frac{\lambda}{k+1}.$$

E.8.3 Descriptive statistics

$$\bar{x}=\frac{1}{n}\sum_{i=1}^{n}x_i, qquads=\sqrt{\frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}}.$$

Weighted mean:

$$\bar{x}_w=\frac{\sum_i w_i x_i}{\sum_i w_i}.$$

Sample proportion and approximate standard error:

$$\hat{p}=\frac{x}{n},\qquad SE(\hat{p})=\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Standard score:

$$z=\frac{x-\mu}{\sigma}.$$

E.8.4 Regression

$$\hat{y}=mx+b, qquade_i=y_i-\hat{y}_i,$$

$$m=\frac{n\sum xy-(\sum x)(\sum y)}{n\sum x^2-(\sum x)^2},\qquad b=\bar{y}-m\bar{x},$$

$$r=\frac{n\sum xy-(\sum x)(\sum y)}{\sqrt{[n\sum x^2-(\sum x)^2][n\sum y^2-(\sum y)^2]}}.$$

Common curve forms:

$$\text{exponential: } \hat{y}=be^{mx},\qquad \text{power: } \hat{y}=bx^m,\qquad \text{logarithmic: } \hat{y}=m\ln x+b.$$

E.8.5 Reliability, capability and decisions

$$R_{\text{series}}=\prod_i R_i, qquadR_{\text{parallel}}=1-\prod_i(1-R_i).$$

$$C_p=\frac{USL-LSL}{6\sigma},\qquad C_{pk}=\min\left(\frac{USL-\mu}{3\sigma},\frac{\mu-LSL}{3\sigma}\right).$$

$$E[C(q,D)]=\sum_d C(q,d)P(D=d).$$

E.9 Verified answer index

These values are collected for quick checking. Extra digits are shown so that later calculations can be reproduced; round only when reporting.

Quantity	Verified value
Diagnostic posterior probability	0.1610169492
Warranty expected cost	R33.90
Binomial $P(X \geq 2)$, $n = 20$, $p = .025$	0.0882417145
Communications $P(X \geq 9)$, $n = 10$, $p = .98$	0.9838223593
Hypergeometric at least one defective	0.3530395137
Hypergeometric exactly two defectives	0.0429874077
Poisson $P(X \geq 4)$, $\lambda = 2.4$	0.2212770890
Normal shaft yield, reference calculation	0.9872757866
PHI program approximation	0.9872757592
Resistor mean	10.005
Resistor sample s	0.0206705764
Weighted production mean	10.007
Shaft mean, all values	20.0030666667
Shaft sample s, all values	0.0120917130
Shaft mean, unusual value excluded	20.0003571429
Shaft sample s, unusual value excluded	0.0062339905
Proportion standard error	0.0180416186
Calibration slope	501.4666666667
Calibration intercept	0.7777777778
Calibration correlation	0.9999973548
Calibration prediction at $x = 2.75$	1379.8111111111
Calibration inverse estimate at $y = 1800$	3.5879198795
RC exponential slope	−0.3159768116
RC coefficient	5.0142326907
RC time constant	3.1647891979 s
RC forecast at $t = 2.5$	2.2758153368 V
Pendulum exponent	0.4993750642
Pendulum coefficient	2.0056125037
Gravity estimate	9.8144435116 m/s^2
Sound logarithmic slope	4.3223052242
Sound intercept	79.7814413430
Delivery slope	1.4566764337 min/km
Delivery intercept	11.5711594810 min
Delivery prediction at 20 km	40.7046881540 min
Series reliability	0.945847
Two-unit parallel reliability	0.9936
3-of-4 voting reliability	0.9477
Minimum inventory expected cost	R123.80 at $q = 3$
C_p	0.8333333333
C_{pk}	0.8055555556

E.10 A reusable reporting page

Copy this page into a notebook or report. A complete record is more useful than a bare display value months later.

Question and context

Decision or question _____

Population or process _____

Event or response _____

Predictor, if any _____

Units and interval _____

Data and method

Source and date _____

Sample size _____

Excluded values and why _____

Method or model _____

Assumptions _____

Calculator settings _____

Result and checks

Expression _____

Unrounded result _____

Reported result _____

Independent check _____

Plain-language finding _____

Limitations _____

Decision or next step _____

What to carry forward

A defensible calculation records not just what buttons produced a number, but what the number means, what assumptions support it, and what decision it is meant to inform.

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Colophon

Typeset in LaTeX using Palatino and Helvetica. All graphs are generated from the data printed in this book. The numerical results and calculator procedures in this edition were independently checked, and every page of the final PDF was visually inspected.